

**CT104-3- PATTERN RECOGNITION****INTAKE: APUMP2601 (Part Time)****LECTURER: ASSOC. PROF. DR. V. SIVAKUMAR****1.0. LEARNING OUTCOMES:**

CLO1	Demonstrate pattern recognition algorithms that can be used to study real-world data (A3, PLO9)	Assignment (Design)
CLO2	Adapt both supervised and unsupervised classification methods to detect and characterize patterns in real-world data (P6, PLO3)	Assignment (Implementation)
CLO3	Analyse the performance of machine learning algorithms for pattern recognition (C4, PLO2).	Assignment (Demo)

COURSE ASSESSMENT**1.1 ASSIGNMENT (DESIGN AND IMPLEMENTATION) – Group Component****1.1.1 ASSIGNMENT INTRODUCTION:** THIS ASSIGNMENT CARRIES 70% OF YOUR TOTAL ASSESSMENT GRADE.

This assignment covers various aspects of pattern recognition from data pre-processing to model evaluation, encouraging students to implement a complete system and document their approach.

SUBMISSION DUE DATE: 27th March, 2026**1.1.2 TASK: Design and Implementation (Development of a Pattern Recognition Application)**

Design and implement any one pattern recognition system chosen from the following.

1. Spam Email Filtering (Classify emails as "Spam" or "Not Spam").
2. Handwritten Digit Recognition (Classify images of handwritten digits (0-9))
3. Predict a medical condition (e.g., Benign/Malignant tumour, Disease/No-Disease) based on a set of patient features (e.g., age, blood pressure, cholesterol levels, etc.).
4. Facial Expression Recognition-Emotion Detection (Classify a face in an image into an emotion (Happy, Sad, Angry, Neutral)).
5. Music Genre Classification (Classify audio clips into genres (Jazz, Classical, Rock, etc.)).

1.2 Goal:

The goal is to develop a pattern recognition system for a chosen application domain, capable of accurately identifying and classifying the relevant patterns.

1.3 Dataset Selection and Preparation:

Gather a suitable dataset relevant to the chosen pattern recognition application domain. Perform data pre-processing steps such as cleaning, normalization, feature extraction, and transformation as necessary. Split the dataset into training, validation, and testing sets.

1.4 Pre-processing:

Implement pre-processing techniques suitable for the chosen pattern recognition application domain. This may involve normalization, colour correction, and other image enhancement techniques.

1.5 Feature Extraction:

Select and implement appropriate feature extraction methods tailored to pattern recognition. This could include texture analysis, colour-based features, or other relevant techniques.

1.6 Model Selection:

Choose a machine learning model suitable for the chosen pattern recognition application domain. Options may include: (the list is not limited...)

- Implementing a **Naive Bayes Classifier**
- Training classifiers like **k-Nearest Neighbors (k-NN)**
- Implementing **Logistic Regression** or **LDA** as a linear classifier.
- Random Forest for feature-based analysis.
- Support Vector Machines (SVM) for pattern detection.

1.7. Training:

- Split the dataset into training and testing sets.
- Train your selected model using the training set.

1.8 Testing and Evaluation:

Evaluate the performance of your model using the testing set. Measure key metrics such as accuracy, sensitivity, specificity, and precision.

1.9 Deployment and Integration:

Deploy the trained model into a functional pattern recognition system. Integrate the system with any necessary infrastructure, interfaces, or user interfaces for practical use in real-world scenarios. Ensure scalability, efficiency, and reliability of the deployed system.

1.10 Optimization:

Experiment with optimization techniques to improve the performance of your pattern recognition system.

1.11 Implementation Documentation:

Prepare a comprehensive report documenting the entire development process, including problem formulation, data collection, model selection, training methodology, evaluation results, and system deployment. Provide detailed insights into the challenges encountered, solutions implemented, and lessons learned during the development process. Include code snippets, visualizations, and diagrams to illustrate key concepts and results effectively.

2.0 ASSIGNMENT (DEMO) – Individual Component

2.1. ASSIGNMENT (DEMO): THIS ASSIGNMENT CARRIES 30% OF YOUR TOTAL ASSESSMENT GRADE

Submission Due Date: 27th March, 2026

2.1.1 TASK:

Creating a demo application for a Pattern Recognition System involves integrating the designed system into a user-friendly interface.

- ❖ An individual demonstration of the completed application.
- ❖ Presentation should be done individually about the developed application by every member of the group and the presentation will last approximately 10-15 minutes.
- ❖ Analyse the system performance.
- ❖ Submit the source code, documentation and a report detailing the results of your pattern recognition system for the developed pattern recognition-based application.
- ❖ All test files (e.g., *.jpg, *.gif, etc.) are to be provided.

2.2. DELIVERABLES:

In addition, a supporting document is to be produced to reflect on the methodologies and implementation tasks applied in the development of the application. In terms of documentation, you are

required to describe and justify the theoretical information about the pattern recognition system developed.

References

All references should be cited within the text and follow APA style.

2.3. DEVELOPMENT TOOLS

❖ Development for the assignment is expected in MATLAB.

However, alternative programming languages, libraries and tools may be used with justification & discussion with the teacher.

2.4. ACADEMIC INTEGRITY

❖ You are expected to maintain the utmost level of academic integrity during the duration of the course.

❖ Plagiarism is a serious offence and will be dealt with according to APU's regulations on plagiarism

2.5. ASSESSMENT CRITERIA

The project will be assessed based on the following key areas:

Criteria	Marks Allotted
Design/Description and justification of the candidate techniques/methods/algorithms with details of the implementation	70%
Prototype/Contribution Demo/Presentation and Critical Analysis	30%